Sri Varshitha Pinnaka

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Education

University of Washington, Seattle, WA — June 2026

B.S. Computer Science and Neuroscience. Robinson Scholar, started university at 15. GPA: 3.84

Relevant Coursework: Machine Learning, Algorithms, Data Structs and Parallelism, Data Visualization, Systems Neuroscience, Cellular/Molecular Neuroscience, Diseases of the Nervous System, Computational Models for Cognitive Neuroscience, Neuropharmacology, Organic Chemistry, Biochemistry, Physics

Research

Human Photonics Lab, University of Washington — Seattle, WA

Undergraduate Research Assistant, advised by Mark Fauver & Eric Seibel — September 2025 — Current

- Testing virtual hematoxylin and eosin (H&E, commonly used in pathology) conversion techniques to improve breast cancer detection. Working towards rapid, algorithmic diagnosis.
- Designing and evaluating automated image processing methodologies employing color balancing and machine learning–based approaches, including CycleGAN/StainGAN, conditional GANs such as StainNet and pix2pix, and LLM-guided virtual H&E conversion.
- Collaborating with interdisciplinary team of medical pathologists, engineers, and researchers across multiple universities to improve CoreView device for accessible, point-of-care breast cancer detection.

Allen Institute for Brain Science — Seattle, WA

Research Intern, advised by Matthew Schmitz — Summer 2024

- Investigated the diversity of brain cell types and evolutionary relationships using deep learning architectures (Enformer) based on CNNs/transformers.
- Implemented Enformer to identify important sequences for transcription regulation within Allen Institute genomes, compared similarity scores across related species, their genomes, and between brain regions of the same species.
- Presented work at annual Allen Institute intern poster symposium.

Molecular Information Systems Lab, UW Paul G. Allen School — Seattle, WA

Undergraduate Research Assistant, advised by Gwendolin Roote, Tracy Mallette, and Zoe Derauf — Oct 2022 — June 2025

- Contributing to molecular biology and computation research, working 15 hours/week. Built leakless strand displacement systems under the Molecular Programming Group, focused on engineering G-quadruplex reporters.
- Recreated traditional semiconductor circuit logic using cellular signaling mechanisms and signaling molecules in bacteria.
- Custom parts assembled and designed for a 'bioprinter' to print spatially separated circuits made of programmable biomolecules.
- Independently ran assays, including PCR, gel electrophoresis, and other DNA quantification/analysis techniques for projects involving bioprinting with the FRESH printing method and Cas9 similarity search.
- Designed scripts using OpenTrons Python API to automate lab tasks.

Other Research

Bruchas Lab, University of Washington — Seattle, WA

Undergraduate Research Assistant, advised by Jingyi Chen — Jan 2023 — Oct 2023

- Conducted random forest and PCA tests to determine the feature that causes stress responses in mice after exposure to chronic isolation and exercise.
- Aligned single-cell imaging data collected with a miniature microscope (using calcium imaging) across trials and days.
- Designed MATLAB and Python algorithms to analyze behavioral event data.

Stuber Lab, University of Washington — Seattle, WA

Undergraduate Research Assistant, advised by Adam Gordon-Fennell — April 2022 — Sept 2022

- Worked 12-15 hours per week contributing to wet-lab neuroscience research. Used fiber photometry, 2-photon spectroscopy, behavioral experiments, and brain histology to understand the mechanism behind consummatory behaviors in the brain.
- Gained experience with 3D printing, circuitry, and assembly of hardware to build behavioral experimental set-ups.

Publications & Presentations

Articles

Virtual Study Abroad and the Meaning of Mobility: A Case Study of the Greece Program at a Large American Public University During the Covid-19 Era

Taso G. Lagos, Anamaria Tepordei, Daniel Chen, **Sri Varshitha Pinnaka**, Alyssa Jones

Journal of Tourism History

https://doi.org/10.1080/1755182X.2026.2678846

Talks

Spatially Separated Computation in Synthetic Biology: Field Programmable Cellular Arrays.

Sri Varshitha Pinnaka. UW Undergrad Research Symposium, May 2024.

Posters

Virtual Hematoxylin and Eosin Colorization for Rapid Point-of-Care Breast Cancer Diagnosis

Sri Varshitha Pinnaka, Eric Seibel, Mark Fauver. UW Undergrad Research Symposium, May 2026.

G-Quadruplex Reporters for Accessible Point of Care Detection.
Sri Varshitha Pinnaka, Sophia Contos, Zoe Derauf, Tracy Mallette, Chris Thachuk. UW Undergrad Research Symposium, May 2025.

Deep learning architectures for investigating enhancer differences across regions and evolution.
Sri Varshitha Pinnaka, Matthew Schmitz, Nelson Johansen. Allen Institute Intern and Postbac Poster Symposium, Aug 2024.

Conference Abstracts

Decoding Claustrum Neuropeptidergic Control of Stress-Induced Binge Eating December 2023
Jingyi Chen, Leandra Mangieri, Sophia Mar, Sean Piantadosi, **Varshitha Pinnaka**, Phoenix Davis, Benjamin Land, Michael Bruchas
American College of Neuropsychopharmacology 62nd *Annual Meeting*
https://doi.org/10.1038/s41386-023-01755-5

Leadership and Campus Involvement

UW Undergraduate Neuroscience Program University of Washington
Teaching Assistant Sept 2025 — Present
- Teaching assistant for python class (NEUSCI 440), helping teach neuroscience majors python for data analysis. Giving lectures, grading quizzes, and answering questions as they arise.
- Guided students, answering questions, and helping with technical issues during weekly upper-level neuroscience lab courses (NEUSCI 301) as part of core teaching team in fall quarter.
- Checking students at vital checkpoints during lab, ensuring they understand underlying biology and analyses for experimental design
- Coordinating with lecturers and lab instructors to alter weekly lab manuals

UW Society of Asian Scientists and Engineers University of Washington
Senior Advisor June 2025 — Present
- Guiding rising SASE leadership team in taking over planning & coordinating of UW SASE events. Oversee transition to new co-presidents and provide feedback based on prior experiences with UW SASE.

Co-President June 2023 — June 2025
- Planning and presenting at monthly events with companies (approx. 40 attendees per event) to help students learn more about STEM.
- Oversaw planning of STEM industry events and working on initiative to increase biotech and science industry presence on campus.
- Oversaw planning of social events to bring together engineering and science events at UW, including bowling, mixers, and game nights.
- Representing SASE at the Engineering Student Council, formed by the UW College of Engineering.

Public Relations Lead June 2022 — June 2023
- Coordinated events with industry partners, including email communications and event logistics. Events were planned to help students learn about innovation and opportunities in various STEM fields.

General Leadership Officer September 2021 — June 2022
- Trained under established officers and learned about company/community outreach, professional event planning, and graphic design.

Volunteering and Shadowing

Seattle Children's Hospital Seattle, WA
Playroom Volunteer April 2025 — Present
- Washed toys, organized toys, and delivered toys to nurses or patients on the units.
- Assisted staff with patients and/or siblings who visit the playroom.

ROOTS Homeless Shelter Seattle, WA
Lead Volunteer July 2026 — Present
- Perform guest intakes: orient new guests to shelter, collect personal details, and connect new guests with resources/case management.
- Assist office staff in scheduling case management appointments, laundry, charging phones, and any other misc. tasks.

Evening Volunteer February 2024 — Present
- Engaged with guests to socialize and build community while working to ensure the emotional and physical safety of the shelter space.
- Set up the shelter each night, handed out hygiene supplies and bedding, and supervised guest activities such as computer time and smoke breaks. Spent time with guests chatting over dinner, playing board games, and watching TV together.

Medical Shadow Virginia Beach, VA
Sentara Virginia Beach General Hospital August — September 2024
- Shadowed physicians in neurosurgery, cardiology, internal medicine, and emergency medicine for 58 hours throughout two weeks.

Jamestown Family Health Clinic March 2023
- Shadowed a variety of providers and specialists for 36 hours throughout one week at the Jamestown Family Health Clinic in as part of the UW Health Care Alternative Spring Break program learning more about rural medicine.

Awards & Honors

Mary Gates Research Scholarship University of Washington
$5,000 to support undergrad research proposal under Human Photonics Lab for CoreView's virtual H&E conversion. December 2025

Jian and Naxin Zhang Undergraduate Student Support Fund Paul G. Allen School, University of Washington
Associated with UW's CSE departmental scholarships, $1,500 awarded for merit. July 2025

CSE Award for Excellence Paul G. Allen School, University of Washington
This $1000 award is merit-based, established by funds from anonymous donors to the Paul G. Allen School. September 2024

Phi Beta Kappa Scholar University of Washington, Seattle
The oldest and nationally most respected academic honors society, recognizes distinguished students in the arts and sciences. July 2024

Skills

- **Programming:** Java, Python, C++, MATLAB, SQL, JavaScript, TypeScript, HTML, CSS
- **Spoken Languages:** English, Spanish, and Telugu